WORKED SOLUTION

STEP III 2015, Question 11

Mechanics • Circular Motion

STEP III 2015, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q11

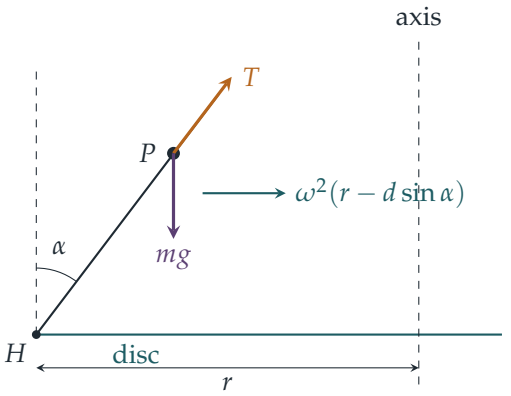
The key idea

A light rod has no mass, so the forces and moments acting on it must balance exactly, even though it is moving. In part (i) only two forces act on the rod, and the one from the hinge has no moment about the hinge, so the force from P must point straight at the hinge. After that, P is a particle moving in a horizontal circle, and resolving does the rest. In part (ii) the forces from the two particles need not lie along the rod, but taking moments about the hinge still gives a single equation for β.

Solution

Model. The particles are point masses, the rod is light and rigid, and the hinge is smooth, so it exerts no couple about its own axis (the horizontal line through the hinge perpendicular to the vertical plane containing the rod). Take axes in that plane, with origin at the hinge H, x horizontal pointing towards the axis of rotation, and y vertically upwards. A point of the rod at distance s from H is at $(s\sin\theta,\ s\cos\theta)$, where θ is the rod's angle with the upward vertical, and it is at horizontal distance $r - s\sin\theta$ from the axis.

Part (i)



The forces on P are drawn above: its weight, and the force from the rod, which we are about to show acts along the rod. The teal arrow is the acceleration of P.

The force from P is along the rod. Only two forces act on the rod: the force from the hinge, acting at H, and the force **G** from P, acting at P. The rod is light, so the total moment on it about the hinge axis is zero (mass times angular acceleration is zero), whatever its motion. The hinge force has no moment about H, so neither does **G**.

Next, P moves in a horizontal circle centred on the axis, so its acceleration is horizontal and points towards the axis, which lies in the vertical plane of the rod. Its weight is in that plane too, so the force on P from the rod, and hence **G** (equal and opposite, by Newton's third law), also lies in that plane. A force in the plane with zero moment about H has its line of action through H. That line also passes through P, which is on the rod, so **G** is parallel to the rod, as required.

The angle. So the rod pushes or pulls P along the rod with a force T, which we take to point from H towards P, i.e. components $(T\sin\alpha,\ T\cos\alpha)$. P is at distance $r - d\sin\alpha$ from the axis, which is positive since $d\sin\alpha < r$, so P is on the same side of the axis as H and its acceleration $\omega^2(r - d\sin\alpha)$ points in the $+x$ direction. Newton's second law for P:

$$\begin{aligned} \text{vertically:} \quad & T\cos\alpha - mg = 0, \\ \text{horizontally:} \quad & T\sin\alpha = m\omega^2(r - d\sin\alpha). \end{aligned}$$

Dividing the second by the first (we have $\cos\alpha \neq 0$, since otherwise $T\cos\alpha = 0 \neq mg$):

$$\tan\alpha = \frac{\omega^2(r - d\sin\alpha)}{g} = \frac{r - d\sin\alpha}{a}.$$

Multiplying by $a\cot\alpha$ gives $a = r\cot\alpha - d\cos\alpha$, that is,

$$r\cot\alpha = a + d\cos\alpha,$$

as required.

The hinge force. The first equation gives $T = mg/\cos\alpha > 0$, so the rod pushes P away from the hinge: the rod is in compression. By Newton's third law P pushes the rod along the rod towards H with force T. The rod is light, so the forces on it sum to zero, and the hinge force on the rod is equal and opposite: it acts **along the rod, away from the hinge towards** P (upwards, and inwards towards the axis), with magnitude

$$\frac{mg}{\cos\alpha} = mg\sec\alpha.$$

Part (ii)

Now the particle of mass m_i $(i = 1, 2)$ is at $(d_i\sin\beta,\ d_i\cos\beta)$, at distance $r - d_i\sin\beta > 0$ from the axis.

> **Ask yourself** Each particle now pulls on the rod in some direction we don't know. How can we avoid finding these forces individually?

We don't need them individually: Newton's second law for each particle gives the force the rod exerts on it, and the moment equation for the light rod about H then removes the unknown hinge force. Let $\mathbf{F}_i$ be the force on particle i from the rod. Its acceleration is $\omega^2(r - d_i\sin\beta)$ in the $+x$ direction, so

$$\mathbf{F}_i + (0, -m_ig) = (m_i\omega^2(r - d_i\sin\beta),\ 0) \quad\Longrightarrow\quad \mathbf{F}_i = (m_i\omega^2(r - d_i\sin\beta),\ m_ig).$$

By Newton's third law, particle i exerts $-\mathbf{F}_i$ on the rod, at the point $(x_i, y_i) = (d_i\sin\beta,\ d_i\cos\beta)$. The anticlockwise moment about H of a force (X, Y) acting at (x, y) is $xY - yX$, so the moment of $-\mathbf{F}_i$ is

$$-d_i\sin\beta\cdot m_ig + d_i\cos\beta\cdot m_i\omega^2(r - d_i\sin\beta).$$

The hinge force has no moment about H, and the rod is light, so these two moments sum to zero. Dividing by $\omega^2\sin\beta$ (which is non-zero: if $\beta = 0$ the moment equation would read $\sum m_id_i\omega^2 r = 0$, which is false) and using $g/\omega^2 = a$:

$$\sum_{i=1}^{2} m_id_i(-a + r\cot\beta - d_i\cos\beta) = 0,$$

that is,

$$(m_1d_1 + m_2d_2)\,r\cot\beta = (m_1d_1 + m_2d_2)\,a + (m_1d_1^2 + m_2d_2^2)\cos\beta.$$

Dividing by $m_1d_1 + m_2d_2 > 0$,

$$r\cot\beta = a + b\cos\beta, \qquad b = \frac{m_1d_1^2 + m_2d_2^2}{m_1d_1 + m_2d_2}.$$

Checks

- With one particle ($m_2 = 0$, say) we get $b = d_1$, recovering part (i). If $d_1 = d_2 = d$, then $b = d$ too: two particles at the same point act like one.

- b is a weighted average of d_1 and d_2 (with weights $m_i d_i$), so it lies between them, as it should.
- As $\omega \to 0$, $a \to \infty$, so $\cot\alpha \to \infty$ and the rod tends to the vertical, as we would expect when the disc barely turns. As $\omega \to \infty$ with $d > r$, $a \to 0$ and the equation tends to $r = d\sin\alpha$: the particle approaches the axis, where it needs almost no centripetal force.
- With $r = 1\,\text{m}$, $\omega = 5\,\text{rad}\,\text{s}^{-1}$, $d = 0.6\,\text{m}$ and $g = 9.81\,\text{m}\,\text{s}^{-2}$, we have $a = 0.3924\,\text{m}$ and the equation gives $\alpha \approx 0.9251$ (about 53.0°); then $r\cot\alpha \approx 0.7538 \approx a + d\cos\alpha$. A computer simulation of the rod's motion started at rest at this angle stays there.

Common slips

- In part (ii), assuming each particle's force on the rod is along the rod. That was special to a single particle; with two, only the total moment about H is zero.
- Using radius r (or $d\sin\alpha$) for the particle's circle instead of $r - d\sin\alpha$.
- Giving the hinge force as acting towards the hinge: the rod is in compression, so the hinge pushes the rod towards P.

Beyond STEP

The moment equation in part (ii) is the condition that the effective potential $\sum m_i\left(g d_i\cos\theta - \frac{1}{2}\omega^2(r - d_i\sin\theta)^2\right)$ is stationary: in the rotating frame, the centrifugal force behaves like an extra potential. This is the starting point for analysing rotating systems such as the centrifugal governor, and for asking whether a steady state is stable. Here the effective potential has a maximum at $\theta = \alpha$, so the steady state of part (i) is in fact unstable: nudge the rod slightly and it swings away. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body moments and rotating frames.

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